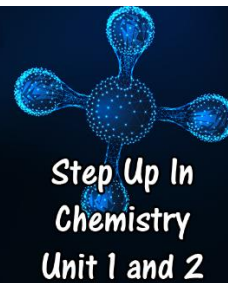


3.3 Reactions of Hydrocarbons

Problems Worksheet



1. Write equations for the combustion of the following hydrocarbons:
 - a. Octane

 - b. Benzene

2. Write equations for the following substitution reactions. Assume each occurs in the presence of sunlight.
 - a. Ethane gas with chlorine gas

 - b. Butane gas with bromine water

 - c. Methane with excess chlorine gas

 - d. Cyclopentane liquid with fluorine gas

3. For each of the following reactions, write an equation and the expected observation for each reaction.
- Ethene gas and chlorine gas
 - Cyclohexene liquid with bromine water
4. For each of the following reactions, write an equation, then draw and name the main organic product of the reaction.
- But-1-ene gas is mixed with hydrogen chloride gas.
 - Hex-1-ene liquid is mixed with hydrogen bromide solution.

5. Write an equation to show how the following compounds could be produced with benzene liquid as a starting material.
- Chlorobenzene
 - Bromobenzene
6. Describe how the following compounds could be produced (no equation necessary).
- Chloroform (trichloromethane)
 - 2-bromoheptane
 - 1,2-dichlorocyclohexane